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# Magnetically Controlled Heat Transfer in a Porous Ferrofluid Enclosure: A Numerical Study for Sustainable Energy Thermal Management

M. J. Sarmadi*, M. R. Dadashzadeh, S. M. Emadi, A. Yazdi

Department of Mechanical Engineering, Babol Noshirvani University of Technology, 47148-71167 Babol, Iran

* Correspondence: M. J. Sarmadi (javadsarmadi@stu.nit.ac.ir)

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**Abstract:** Thermal management is essential for the efficient and reliable operation of energy systems, particularly where heat transfer must be regulated under varying thermal conditions. Magnetically responsive fluids offer a means of controlling convective transport without mechanically altering the flow domain. However, the combined effects of spatially non-uniform magnetic fields, porous-medium permeability, thermal radiation, and nanoparticle concentration on heat transfer in geometrically complex enclosures remain insufficiently understood. This study investigates the coupled thermal and hydrodynamic behavior of $Fe_3O_4$–water ferrofluid in a porous enclosure to examine its potential for controllable thermal management in energy-related applications. A two-dimensional numerical model was formulated for an enclosure with an inner heated elliptical boundary and an outer sinusoidal cooled wall. The governing equations were solved using the control volume finite element method (CVFEM), and the influences of the Hartmann number ($Ha$), Rayleigh number ($Ra$), Darcy number ($Da$), radiation parameter ($Rd$), and nanoparticle volume fraction ($\varphi$) were examined. The numerical predictions were compared with published results, and a mesh-independence assessment was performed. Increasing the magnetic field strength reduced the average Nusselt number by up to approximately 29.72%, indicating suppression of buoyancy-driven convection. In contrast, increasing the $Ra$ raised the Nusselt number by up to 150.22% in the absence of magnetic forcing. Higher porous-medium permeability increased the Nusselt number by up to 4.58%, while increasing the $Rd$ produced an increase of up to 104.75%. Raising the $Fe_3O_4$ $\varphi$ increased the Nusselt number by up to 5.98%. These results indicate that magnetic forcing provides a means of regulating convective heat transfer, while permeability, radiation, and nanoparticle loading influence the thermal response of the enclosure. The study identifies the competing physical mechanisms governing ferrofluid heat transfer and provides a basis for further investigation of magnetically controlled thermal components in sustainable energy systems.

**Keywords:** Sustainable energy thermal management; Magnetically controlled heat transfer; $Fe_3O_4$ ferrofluid; Porous enclosure; Natural convection; Thermal radiation; Numerical simulation

## 1 Introduction

Thermal management remains an important consideration in the development of sustainable energy technologies. The performance and operational reliability of solar thermal systems, thermal energy storage units, compact heat exchangers, and other energy-related devices depend partly on the control of heat transfer and temperature distribution. In systems exposed to changing thermal loads, the ability to regulate thermal transport can be as important as increasing the heat transfer rate. The use of magnetically responsive working fluids with Hartmann numbers ($Ha$) has attracted attention because externally applied magnetic fields can influence fluid circulation without requiring mechanical changes to the flow domain. Mehrez et al. [1] reviewed the application of ferrofluids and magnetic fields in solar energy systems, discussing their thermal, rheological, and optical characteristics as well as the challenges associated with practical implementation. Peer et al. [2] examined nanofluids in thermal energy storage systems and emphasized the importance of thermophysical properties and operating conditions in determining their suitability for energy applications. These studies indicate that magnetically responsive fluids merit further investigation as

potential working media in controllable thermal components, although improvements in component heat transfer do not necessarily translate directly into higher system efficiency.

Magnetohydrodynamic transport has received considerable attention in thermal engineering because magnetic fields can modify fluid motion and heat transfer in electrically conductive media. The interaction between an applied magnetic field and the induced electric current generates Lorentz forces that influence velocity distributions, circulation patterns, and convective transport. The resulting response depends on the electrical properties of the fluid, magnetic field configuration, and prevailing flow conditions. Uniform magnetic fields are commonly considered in numerical investigations because of their relatively simple spatial distributions. Non-uniform fields introduce position-dependent electromagnetic effects and can alter local circulation patterns. In buoyancy-driven systems, the magnetic contribution may oppose or redistribute fluid motion, depending on the field orientation, strength, and adopted physical model. Such effects have motivated investigations of magnetic regulation in confined thermal systems [3–5]. Huang et al. [6] reviewed coupled thermal, fluid, and magnetic mechanisms in composite phase change systems, distinguishing between magnetohydrodynamic and ferrohydrodynamic effects. Their analysis also highlighted the importance of identifying the physical mechanisms responsible for magnetic control rather than treating all magnetic interactions as equivalent.

Ferrofluids provide another means of modifying thermal transport through the dispersion of magnetic nanoparticles in a carrier liquid. Water-based suspensions containing $Fe_3O_4$ nanoparticles are of particular interest because their effective thermophysical properties differ from those of the base fluid and their response to an external magnetic field may influence fluid circulation. Increasing nanoparticle concentration can raise effective thermal conductivity, but the resulting convective response also depends on viscosity, density, and the prevailing flow regime. Consequently, an increase in thermal conductivity does not necessarily produce a corresponding increase in natural convection. These competing effects are important when considering ferrofluids for thermal management systems operating under variable conditions [7–9]. Mirahmad et al. [10] reviewed the influence of nanoparticle composition, concentration, specific heat capacity, and fluid stability on thermal energy storage and heat transfer. Their findings underline the need to evaluate nanoparticle loading in terms of several thermophysical properties rather than thermal conductivity alone.

Porous structures introduce additional interactions between fluid motion and heat transfer. The permeability and geometric characteristics of a porous matrix determine the resistance encountered by the fluid and influence the velocity and temperature distributions within an enclosure. Depending on the material properties and thermal boundary conditions, a porous medium may facilitate thermal exchange or restrict convective circulation. Such structures are used in geothermal heat extraction, thermal storage components, compact heat exchangers, and insulation systems. When a magnetically responsive fluid occupies a porous domain, the resistance imposed by the matrix interacts with buoyancy and magnetic forcing, making the resulting flow behavior dependent on more than permeability alone. Previous investigations have examined the combined influence of magnetic fields, porous resistance, and nanofluid properties on heat transfer in confined geometries [11–14].

Recent studies have examined these interactions using different porous-medium and magnetic models. Ayadi et al. [15] investigated natural convection in a partitioned porous enclosure subjected to permanent magnetic fields, considering the interaction between buoyancy-driven and thermomagnetic convection. Hajialiloo and Soltanipour [16] studied ferrofluid convection in a porous cavity using a local thermal non-equilibrium formulation. Their results showed that magnetic source location, field configuration, permeability, and the thermal interaction between the fluid and solid matrix can affect the heat transfer response. These studies demonstrate that porous ferrofluid systems may exhibit different thermal behavior depending on the adopted magnetic and porous-medium assumptions. They also emphasize the importance of distinguishing simplified homogeneous representations from models that separately resolve the thermal behavior of the fluid and porous matrix.

Thermal radiation introduces a further contribution to heat transport, particularly in configurations where radiative exchange is appreciable relative to conduction and convection. Its inclusion in the energy equation changes the temperature distribution and may influence the thermal gradients responsible for buoyancy-driven circulation. The relative contribution of radiation also affects the interpretation of heat transfer indicators when the adopted Nusselt number formulation includes a radiative component. From the perspective of thermal management, this distinction matters because an increase in the reported heat transfer indicator does not necessarily represent an equivalent increase in convection. The combined influence of radiation, magnetic forcing, and porous resistance therefore requires consideration within the physical assumptions of the numerical model.

Earlier studies provide a basis for understanding the effects of geometry, buoyancy, and nanoparticle loading on enclosure convection. Azizul et al. [17] investigated heatline distributions in a nanofluid-filled enclosure with a sinusoidal heated boundary and an internal solid body. Their results illustrated the influence of Reynolds and Grashof numbers on fluid circulation and heat transfer behavior. Li and Peterson [18] experimentally examined natural convection in $Al_2O_3$–water nanofluids and reported that increasing nanoparticle concentration could reduce heat transfer under the investigated conditions. El Abdallaoui et al. [19] numerically studied natural convection

around a triangular heated cylinder using the lattice Boltzmann method and examined the influence of geometric confinement and nanoparticle concentration. Together, these investigations show that the convective response of nanofluids depends strongly on geometry, particle properties, and thermal conditions.

More recent numerical studies have extended these investigations to configurations involving magnetic forcing, radiation, and complex porous boundaries. Armaghani et al. [20] examined unsteady magnetohydrodynamic natural convection of a hybrid nanofluid in a U-shaped wavy porous cavity, including the influence of thermal radiation. Their analysis showed that magnetic damping, radiative transport, and cavity geometry can substantially alter the heat transfer response. Jakeer et al. [21] investigated entropy generation and heat transfer in a partly heated I-shaped porous enclosure containing hybrid nanofluids under magnetic and radiative effects. Their study considered Rayleigh number ($Ra$), Darcy number ($Da$), $Ha$, nanoparticle concentration, and radiation, demonstrating the close relationship between transport mechanisms and thermal irreversibility. These investigations establish that the simultaneous treatment of magnetic effects, porous media, radiation, and irregular geometries is not itself an unexplored research direction.

The influence of magnetic forcing on natural convection has also been investigated in other enclosure configurations. Oğlakkaya and Bozkaya [22] examined magnetohydrodynamic natural convection in a semi-annular enclosure containing a water-based nanofluid and analyzed the effects of $Ra$ and $Ha$ on fluid motion and heat transfer. Boualit et al. [23] studied natural convection in a square cavity using a nanofluid dispersion model and reported variations in heat transfer associated with nanoparticle size. More recently, Alwan et al. [24] investigated a triangular enclosure containing a hybrid nanofluid and a porous wavy fin under magnetic and radiative effects. Their work demonstrated the importance of buoyancy, magnetic field orientation, radiation, and porous geometry in determining the resulting thermal response. Mousavi et al. [25] reviewed magnetic regulation in phase change thermal storage systems, highlighting the influence of magnetic field characteristics and nanoparticle properties on heat transfer during melting and solidification. Although these storage studies involve different physical processes from the present steady-state enclosure problem, they establish the broader relevance of magnetic control to thermal energy applications.

The available literature shows that magnetic forcing, porous resistance, radiation, and nanoparticle loading have been investigated in several combinations. However, the reported thermal response depends on the enclosure geometry, applied field distribution, boundary conditions, working fluid, and governing physical assumptions [26]. Consequently, results obtained for uniform magnetic fields, conventional cavity geometries, or different porous-medium treatments cannot be transferred directly to configurations subjected to localized magnetic forcing. For a geometrically irregular enclosure, the interaction between buoyancy-driven circulation and spatially varying electromagnetic terms may produce changes in the flow structure that are not adequately described by the influence of individual parameters alone. The relative importance of permeability and radiation under different magnetic conditions also requires careful interpretation, particularly when the reported Nusselt number incorporates more than one heat transfer contribution.

The present study therefore investigates natural convection in a porous enclosure filled with $Fe_3O_4$–water ferrofluid and subjected to a spatially non-uniform magnetic field. The computational domain consists of an inner heated elliptical boundary and an outer sinusoidal cooled wall. The governing equations are solved using the control volume finite element method (CVFEM), and the effects of the $Ha$, $Ra$, $Da$, radiation parameter ($Rd$), and nanoparticle volume fraction ($\varphi$) are examined through changes in fluid circulation, temperature distribution, and average Nusselt number.

The principal contribution is a numerical assessment of these interacting parameters within a single irregular porous enclosure under localized magnetic forcing. Particular attention is given to the competition between magnetic suppression of convection and the thermal responses associated with buoyancy, permeability, radiation, and nanoparticle loading. The analysis is intended to clarify the behavior predicted by the adopted physical model rather than establish the performance of a complete energy conversion or storage device. The results provide a reference for further investigations of controllable ferrofluid heat transfer in thermal components relevant to sustainable energy systems, where both heat transport and the energy required for external regulation must ultimately be considered.

## 2 Problem Definition and Numerical Modeling Approach

The present study considers buoyancy-driven heat transfer in a porous enclosure filled with an $Fe_3O_4$–water ferrofluid and subjected to an externally imposed, spatially non-uniform magnetic field. The numerical model was developed to examine how magnetic forcing interacts with porous-medium resistance, thermal radiation, and nanoparticle concentration in determining fluid circulation and heat transfer. These interactions are relevant to the assessment of controllable thermal components for energy-related applications, where changes in temperature distribution and heat transport may affect system operation. The computational domain represents an idealized thermal configuration rather than a complete energy conversion or storage device. Accordingly, the analysis focuses on the physical mechanisms governing heat transfer within the enclosure, without assuming a direct improvement in

system-level energy efficiency.

The physical configuration consists of a two-dimensional porous enclosure bounded internally by an elliptical heated surface and externally by a sinusoidal cooled wall. The inner elliptical boundary is subjected to a prescribed uniform heat flux, whereas the outer boundary is maintained at a constant lower temperature. This arrangement provides a means of examining heat transfer between irregularly shaped boundaries under buoyancy-driven flow conditions. Such geometries are useful for investigating thermal transport in confined spaces, although the present configuration is not intended to reproduce the geometry of a particular commercial heat exchanger or thermal storage unit.

An external magnetic source is located near the inner elliptical boundary, producing a non-uniform magnetic field within the computational region. Unlike a spatially uniform field, this configuration introduces variations in magnetic intensity across the enclosure and consequently allows the influence of localized magnetic forcing on fluid circulation to be examined. The magnetic contribution was represented using the electromagnetic terms specified in the governing equations. Its physical interpretation depends on the adopted assumptions concerning fluid electrical conductivity and magnetic response.

The working medium is water containing dispersed $Fe_3O_4$ nanoparticles. A homogeneous single-phase approximation was adopted, in which the suspension was represented by effective thermophysical properties determined from the $\varphi$ and the properties of its constituent materials. This treatment allows changes in density, viscosity, thermal conductivity, heat capacity, and electrical conductivity to be incorporated into the numerical model without resolving the motion of individual particles. It does not explicitly describe nanoparticle aggregation, sedimentation, or relative particle migration. These simplifications should be considered when relating the numerical results to practical ferrofluid-based thermal devices.

The effects of the principal governing parameters were examined through variations in $Ha$, $Da$, $\varphi$, $Rd$, and $Ra$. These parameters characterize the modeled electromagnetic influence, porous-medium permeability, nanoparticle loading, radiative transport, and buoyancy-driven convection, respectively. Their combined consideration permits an assessment of the competing effects that determine the thermal response of the enclosure. In particular, the analysis distinguishes changes in convective circulation from changes in the overall heat transfer indicator, an important distinction when evaluating possible thermal management applications.

The physical model and the location of the external magnetic source are illustrated in Figure 1. The thermophysical properties of water and $Fe_3O_4$ used in the calculations are listed in Table 1. The spatial distribution of the imposed magnetic field, obtained from Eqs. (1)–(3), is shown in Figure 2. These expressions define the magnetic field associated with the external source and provide the spatially varying magnetic quantities required by the flow model.

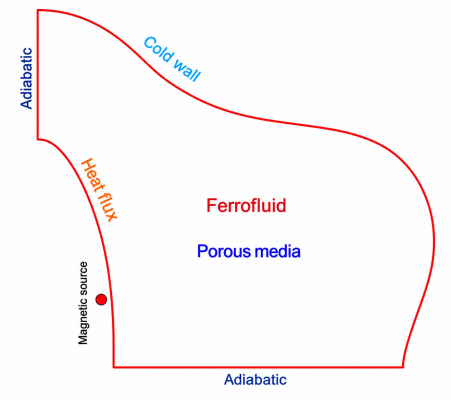
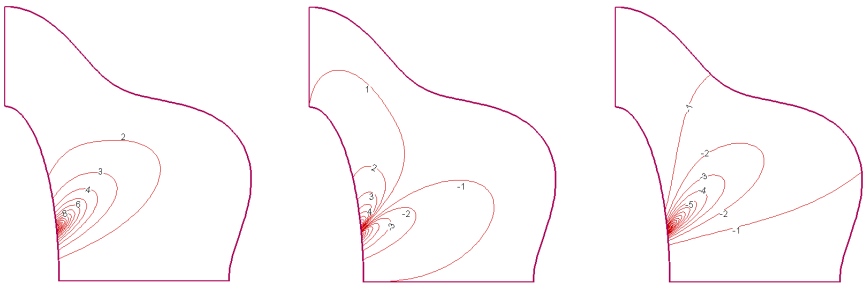



**Figure 1.** Geometry and thermal boundary configuration of the porous enclosure with an externally applied magnetic field

**Table 1.** Thermophysical properties of the base fluid and $Fe_3O_4$ nanoparticles used in the numerical model [27]

| **Property** | **Water** | **$Fe_3O_4$** |
|---|---|---|
| Density, $\rho$ (kg $\cdot$ m$^{-3}$) | 997.1 | 5,200 |
| Specific heat capacity, $C_p$ (J $\cdot$ kg$^{-1}$ $\cdot$ K$^{-1}$) | 4,179 | 670 |
| Thermal conductivity, $k$ (W $\cdot$ m$^{-1}$ $\cdot$ K$^{-1}$) | 0.613 | 6 |
| Thermal expansion coefficient, $\beta$ ($10^{-5}$ $\cdot$ K$^{-1}$) | 21 | 1.3 |
| Electrical conductivity, $\sigma$ (S $\cdot$ m$^{-1}$) | 0.05 | 25,000 |



(a) (b) (c)

**Figure 2.** Spatial distribution of the magnetic field within the computational domain: (a) $H(x,y)$; (b) $H_x(x,y)$; (c) $H_y(x,y)$

The magnetic field generated by the external wire was described using the prescribed analytical formulation. The horizontal and vertical components of the magnetic field intensity, $\bar{H}_x$ and $\bar{H}_y$ $(\mathrm{A}\cdot\mathrm{m}^{-1})$, were calculated using Eqs. (1) and (2), respectively. The parameter $\gamma$ (A) denotes the electric current flowing through the wire, while $a$ and $b$ (m) specify its position in the computational coordinate system. The magnitude of the magnetic field intensity, $\bar{H}$ $(\mathrm{A}\cdot\mathrm{m}^{-1})$, was determined from Eq. (3).

$$\bar{H}_y = -\frac{\gamma}{2\pi}\frac{1}{(x-\bar{a})^2+(y-\bar{b})^2}(x-\bar{a}) \tag{1}$$

$$\bar{H}_x = \frac{\gamma}{2\pi}\frac{1}{(x-\bar{a})^2+(y-\bar{b})^2}(y-\bar{b}) \tag{2}$$

$$\bar{H} = \sqrt{\bar{H}_x^2+\bar{H}_y^2} = \frac{\gamma}{2\pi}\frac{1}{\sqrt{(x-\bar{a})^2+(y-\bar{b})^2}} \tag{3}$$

The fluid motion and heat transfer were described using the steady-state governing equations [28]. The formulation includes conservation of mass, momentum transport in the horizontal and vertical directions, and thermal energy transport. The equations incorporate buoyancy, porous-medium resistance, electromagnetic source terms, and the modeled contribution of thermal radiation.

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{4}$$

$$\rho_{nf}\left(v\frac{\partial u}{\partial y}+u\frac{\partial u}{\partial x}\right) = -\frac{\partial p}{\partial x}+\mu_{nf}\left(\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial x^2}\right)-uB_y^2\sigma_{nf}+vB_yB_x\sigma_{nf}-\frac{1}{\rho_{nf}}\frac{\mu_{nf}}{K}u \tag{5}$$

$$\begin{gathered}\rho_{nf}\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y}+\mu_{nf}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)-\sigma_{nf}B_x^2v+uB_yB_x\sigma_{nf}-\frac{1}{\rho_{nf}}\frac{\mu_{nf}}{K}u \\ +\rho_{nf}\beta_{nf}g\,(T-T_c) \\ \bar{B}=\mu_0\bar{H},\mu_0=4\pi\times10^{-7}(\mathrm{T}\cdot\mathrm{m/A})\end{gathered} \tag{6}$$

$$\begin{gathered}\frac{1}{(\rho C_p)_{nf}}\frac{\partial q_r}{\partial y}+(C_p\rho)_{nf}\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=k_{rf}\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)+\sigma_{nf}\left(B_xv-B_yu\right)^2 \\ T^4\cong 4T_c^3T-3T_c^4, q_r=-\frac{4\sigma_e}{3\beta_R}\frac{\partial T^4}{\partial y}\end{gathered} \tag{7}$$

In these equations, $u$ and $v$ $(\mathrm{m}\cdot\mathrm{s}^{-1})$ denote the velocity components in the $x$ and $y$ directions, respectively, while $p$ (Pa) represents pressure. The subscript nf refers to the effective properties of the ferrofluid. The parameters $\rho_{nf}$ $(\mathrm{kg}\cdot\mathrm{m}^{-3})$, $\mu_{nf}$ (Pa·s), and $k_{nf}$ $(\mathrm{W}\cdot\mathrm{m}^{-1}\cdot\ \mathrm{K}^{-1})$ are the effective density, dynamic viscosity, and thermal conductivity, respectively.

Eq. (4) represents mass conservation for incompressible flow, while Eqs. (5) and (6) describe momentum transport in the horizontal and vertical directions, respectively. The momentum equations include viscous diffusion, electromagnetic contributions, porous-medium resistance, and buoyancy, as specified in the adopted formulation.

The permeability of the porous medium is denoted by $K$ ($\mathrm{m}^2$), while $\beta_{nf}$ ($\mathrm{K}^{-1}$) represents the effective thermal expansion coefficient. Gravitational acceleration is denoted by $g$ ($\mathrm{m}\cdot\mathrm{s}^{-2}$), and $T_c$ (K) is the prescribed temperature of the cooled boundary. The momentum equations account for the resistance imposed by the porous matrix and the electromagnetic terms included in the adopted formulation. The relative importance of these mechanisms determines the resulting circulation patterns and their contribution to convective heat transfer.

Eq. (7) represents the thermal energy balance. The effective specific heat capacity is denoted by $(C_p)_{nf}$ ($\mathrm{J}\cdot\mathrm{kg}^{-1}\cdot\mathrm{K}^{-1}$), $T$ (K) is temperature, and $q_r$ ($\mathrm{W}\cdot\mathrm{m}^{-2}$) represents radiative heat flux. The energy equation includes the modeled electromagnetic dissipation contribution and the radiative transport term. Thermal radiation was treated using the Rosseland approximation, in which $\sigma_e$ ($\mathrm{W}\cdot\mathrm{m}^{-2}\cdot\mathrm{K}^{-4}$) is the Rosseland mean extinction coefficient and $\beta_R$ ($\mathrm{m}^{-1}$) is the Rosseland mean extinction coefficient. The applicability of this approximation depends on the optical characteristics of the medium and the assumptions used to represent radiative transfer.

The inclusion of thermal radiation allows the numerical model to examine changes in temperature distribution arising from an additional energy transport mechanism. For thermal management applications, such changes are of interest because the balance between conduction, convection, and radiation influences the thermal response of a component. However, the contribution of radiation to the calculated Nusselt number must be distinguished from the convective contribution when interpreting the results.

The porous medium was represented through the resistance terms specified in the momentum equations. The mathematical formulation accounts for the effect of permeability on the fluid velocity and the associated convective transport. Magnetic effects were introduced through the electromagnetic terms of the governing equations, while changes in nanoparticle concentration were represented through the effective thermophysical properties of the homogeneous suspension. These properties were calculated using the following relationships [28]:

$$(\rho C_p)_{nf} = (\rho C_p)_s \varphi + (1-\varphi)(\rho C_p)_f \tag{8}$$

$$\rho_{nf} = \rho_f(1-\varphi) + \rho_s\varphi \tag{9}$$

$$\beta_{nf} = \beta_f(1-\varphi) + \beta_s\varphi \tag{10}$$

$$\mu_{nf} = \frac{\mu_f}{(1-\varphi)^{2.5}} \tag{11}$$

$$\frac{\sigma_{nf}}{\sigma_f} = 1 + \frac{3\left(-1+\frac{\sigma_s}{\sigma_f}\right)\varphi}{-\left(-1+\frac{\sigma_s}{\sigma_f}\right)\varphi + \left(2+\frac{\sigma_s}{\sigma_f}\right)} \tag{12}$$

$$\frac{k_{nf}}{k_f} = \frac{-2\varphi(k_f-k_s)+k_s+2k_f}{\varphi(k_f-k_s)+k_s+2k_f} \tag{13}$$

The subscripts $f$, $s$, and $nf$ denote the base fluid, solid nanoparticles, and effective nanofluid properties, respectively. The nanoparticle volume fraction is represented by $\varphi$. Eqs. (8) and (9) provide the effective volumetric heat capacity and density according to mixture relationships, while Eq. (10) defines the effective thermal expansion property. The effective dynamic viscosity was evaluated using the Brinkman model in Eq. (11). Eqs. (12) and (13) describe the effective electrical and thermal conductivities using the adopted mixture-based property models.

These relationships provide a consistent means of varying the effective material properties with nanoparticle concentration within the assumptions of the single-phase model. Nevertheless, they do not account for all mechanisms present in real ferrofluid suspensions, including temperature-dependent aggregation, magnetic-field-dependent microstructure, and non-uniform particle distributions. The calculated influence of nanoparticle loading should therefore be interpreted as the response predicted by the selected effective-property models.

To facilitate numerical analysis and comparison of the governing physical mechanisms, the mathematical formulation was expressed in dimensionless form. The dimensionless variables were defined as follows:

$$\begin{aligned} &a = \frac{\bar{a}}{L}, b = \frac{\bar{b}}{L}, Y = \frac{y}{L}, \Theta = \frac{T-T_c}{T_h-T_c}, U = \frac{uL}{\alpha_f}, V = \frac{vL}{\alpha_f}, \\ &H_x = \frac{\bar{H}_x}{\bar{H}_0}, H = \frac{\bar{H}}{\bar{H}_0}, H_y = \frac{\bar{H}_y}{\bar{H}_0}, X = \frac{x}{L}, P = \frac{p}{\rho_f(\alpha_f/\mathrm{L})^2} \end{aligned} \tag{14}$$

Here, $L$ (m) denotes the characteristic length, $T_h$ and $T_c$ (K) are the reference hot and cold temperatures, $\alpha_f$ ($\mathrm{m}^2 \cdot \mathrm{s}^{-1}$) is the thermal diffusivity of the base fluid, and $\rho_f$ ($\mathrm{kg} \cdot \mathrm{m}^{-3}$) is its density.

Substituting the dimensionless variables into the governing equations yielded the following dimensionless system [29]:

$$\frac{\partial V}{\partial Y} + \frac{\partial U}{\partial X} = 0 \tag{15}$$

$$\begin{aligned} U\frac{\partial U}{\partial X} + V\frac{\partial U}{\partial Y} &= -\frac{\partial P}{\partial X} + \Pr\left[\frac{\mu_{nf}/\mu_f}{\rho_{nf}/\rho_f}\right]\left(\frac{\partial^2 U}{\partial Y^2} + \frac{\partial^2 U}{\partial X^2}\right) \\ -Ha^2 \Pr\left[\frac{\sigma_{nf}/\sigma_f}{\rho_{nf}/\rho_f}\right]\left(H_y^2 U - H_x H_y V\right) &- \frac{\Pr}{Da}\left[\frac{\mu_{nf}/\mu_f}{\rho_{nf}/\rho_f}\right] U \end{aligned} \tag{16}$$

$$\begin{aligned} U\frac{\partial V}{\partial X} + V\frac{\partial V}{\partial Y} &= -\frac{\partial P}{\partial X} + \Pr\left[\frac{\mu_{nf}/\mu_f}{\rho_{nf}/\rho_f}\right]\left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2}\right) \\ -Ha^2 Pr\left(-UH_xH_y + VH_x^2\right)\left[\frac{\sigma_{nf}/\sigma_f}{\rho_{nf}/\rho_f}\right] &+ \left[\frac{\beta_{nf}}{\beta_f}\right] Pr\Theta Ra - \frac{Pr}{Da}\left[\frac{\mu_{nf}/\mu_f}{\rho_{nf}/\rho_f}\right] V \end{aligned} \tag{17}$$

$$\begin{aligned} U\frac{\partial \Theta}{\partial X} + V\frac{\partial \Theta}{\partial Y} &= \left[\frac{\frac{k_{nf}}{k_f}}{\frac{(\rho C_P)_{nf}}{(\rho C_P)_f}}\right]\left(\frac{\partial^2 \Theta}{\partial X^2} + \left(1 + \frac{4}{3}\left(\frac{k_{nf}}{k_f}\right)^{-1} Rd\right)\frac{\partial^2 \Theta}{\partial Y^2}\right) \\ &+ Ha^2 Ec\left[\frac{\frac{\sigma_{nf}}{\sigma_f}}{\frac{(\rho C_P)_{nf}}{(\rho C_P)_f}}\right]\{UH_y - VH_x\}^2 \end{aligned} \tag{18}$$

The dimensionless parameters describe the relative contributions of buoyancy, viscous diffusion, magnetic forcing, and porous resistance. $Ra$ characterizes buoyancy-driven convection relative to the thermal and momentum diffusion processes, while the Prandtl number is defined as $P_r$ = $v_f/\alpha_f$, where $v_f$ ($\mathrm{m}^{2\cdot}\ \mathrm{s}^{-1}$) is the kinematic viscosity of the base fluid. $Ec$ is the Eckert number, $Ha$ characterizes the relative influence of the magnetic field in the adopted electrically conducting fluid model. The $Da$ is defined as $Da$ = $K/L^2$ and represents the dimensionless permeability of the porous medium.

The characteristic magnetic field intensity is denoted by $H_0$ ($\mathrm{A} \cdot \mathrm{m}^{-1}$), while $\sigma_f$ ($\mathrm{S} \cdot \mathrm{m}^{-1}$) and $\mu_f$ ($\mathrm{Pa} \cdot \mathrm{s}$) denote the electrical conductivity and dynamic viscosity of the base fluid, respectively. The characteristic magnetic flux density corresponding to $H_0$ is $B_0$ = $\mu_0 H_0$ under the assumed permeability relation.

The stream-function and vorticity formulation was employed to describe the two-dimensional flow field. The dimensional stream function $\psi$ ($\mathrm{m}^2 \cdot\ \mathrm{s}^{-1}$) and vorticity $\omega$ ($\mathrm{s}^{-1}$) were converted to the dimensionless quantities $\psi$ and $\Omega$ using the definitions provided in Eq. (19). These variables permit the circulation structure to be examined alongside the temperature distribution and facilitate the interpretation of buoyancy-driven and magnetically influenced flow patterns.

The momentum equations were reformulated in terms of the stream function and vorticity according to the numerical approach described in the cited studies [28, 29].

$$\begin{aligned} u = \frac{\partial \psi}{\partial y}, v = -\frac{\partial \psi}{\partial x}, \\ \Omega = \frac{\omega L^2}{\alpha_f}, \Psi = \frac{\psi}{\alpha_f}, \omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \end{aligned} \tag{19}$$

The heat transfer response was quantified using the local and average Nusselt numbers. The local Nusselt number was evaluated along the heated boundary using the adopted dimensionless temperature gradient and the radiation contribution specified in the model. The average Nusselt number was obtained by integrating the local quantity over the heated-wall length.

$$Nu_{loc} = \frac{1}{\theta}\left(\frac{k_{nf}}{k_f}\right)\left(1 + \frac{4}{3}\left(\frac{k_{nf}}{k_f}\right)^{-1} Rd\right) \tag{20}$$

$$Nu_{ave} = \frac{1}{S}\int_0^s Nu_{loc} \mathrm{d}s \tag{21}$$

In these expressions, $Nu_{loc}$ is the local Nusselt number, $\theta$ is the dimensionless temperature, $s$ denotes the coordinate along the heated boundary, and $S$ is the total length of that boundary. Eq. (21) defines the wall-averaged Nusselt number. Because the model includes thermal radiation, the interpretation of this quantity depends on whether the adopted definition represents conductive heat transfer alone or incorporates a radiative contribution.

The average Nusselt number was used to compare the thermal response of the enclosure under different magnetic, buoyancy, permeability, radiation, and nanoparticle-loading conditions. It provides a measure of heat transfer within the specified numerical formulation but should not be interpreted as a direct measure of overall energy efficiency, thermal storage effectiveness, or reductions in operating energy consumption. Establishing such system-level performance would require additional information on device geometry, operating conditions, thermal loads, and energy input.

The governing equations were solved using the CVFEM. This numerical approach combines a finite-element representation of the computational domain with control-volume-based discretization of the conservation equations. It is suited to geometrically irregular domains and permits the governing transport equations and associated source terms to be evaluated within a common computational framework. The approach has previously been applied to numerical investigations of heat transfer in nanofluid systems [29].

A mesh-independence assessment and comparison with published numerical results were used to examine the consistency of the computed solution, as reported in Section 3. The subsequent analysis considers changes in circulation patterns, temperature distributions, and average Nusselt number over the investigated parameter ranges. By examining these quantities together, the numerical study identifies the mechanisms governing magnetically regulated heat transfer in the idealized porous enclosure and provides a basis for further component-level investigations involving thermally responsive working fluids in energy applications.

## 3 Results and Discussion

The numerical results are presented to examine the combined influence of magnetic forcing, buoyancy, porous-medium permeability, thermal radiation, and nanoparticle concentration on heat transfer within the $Fe_3O_4$–water ferrofluid enclosure. Particular attention is given to the relationship between changes in fluid circulation and the corresponding thermal response. The analysis considers a geometrically irregular domain bounded by an inner elliptical heated surface and an outer sinusoidal cooled wall, with a non-uniform magnetic field imposed by an external source positioned near the heated boundary. This arrangement makes it possible to investigate the competition between buoyancy-driven motion, electromagnetic damping, porous resistance, and radiative transport under different operating conditions. From the perspective of sustainable energy technologies, these interactions are relevant to the preliminary assessment of thermally controllable components, where regulating heat transfer and temperature distribution may be necessary under variable thermal loads.

Before examining the effects of the governing parameters, the numerical predictions were evaluated against published results. Figure 3 compares the calculated average Nusselt numbers with the corresponding values reported by Luo et al. [27] under the specified reference conditions. The comparison showed close agreement between the present calculations and the published data, indicating that the numerical implementation reproduced the reference heat transfer behavior for the selected cases. This comparison provides a basis for assessing the consistency of the computational approach, although it does not independently establish the accuracy of every physical mechanism included in the model.

The sensitivity of the numerical solution to grid resolution was also examined. Table 2 presents the average Nusselt numbers calculated using five computational grids under identical operating conditions. The predicted Nusselt number increased from 9.241356 on the coarsest grid to 9.269803 on the finest grid, with progressively smaller differences between successive refinements. The grid containing $71\times211$ elements produced an average Nusselt number of 9.269544, compared with 9.269803 on the finer $81\times241$ grid. The relative difference was approximately 0.0028%, indicating that further refinement had little influence on the calculated heat transfer indicator under the tested conditions. The $71\times211$ grid was therefore selected for the subsequent simulations as a compromise between numerical resolution and computational requirements. The assessment supports grid convergence for the reported Nusselt number but does not, by itself, establish grid independence for all local velocity and temperature gradients.

The effects of magnetic field strength, buoyancy, and porous-medium permeability on the flow structure and temperature distribution are illustrated in Figure 4 and Figure 5. These figures present the streamline and isotherm patterns obtained under different combinations of $Ha$, $Da$, and $Ra$. Increasing $Ha$ strengthened the electromagnetic influence represented in the governing equations and weakened the buoyancy-driven circulation. The streamline patterns showed a reduction in circulation intensity, accompanied by changes in the position and shape of the dominant recirculating region. Under stronger magnetic forcing, the principal vortex became displaced and extended toward the upper part of the enclosure. This behavior is consistent with the damping contribution of the electromagnetic terms in the adopted model, which restricts fluid motion and alters the balance between convective transport and

thermal diffusion.

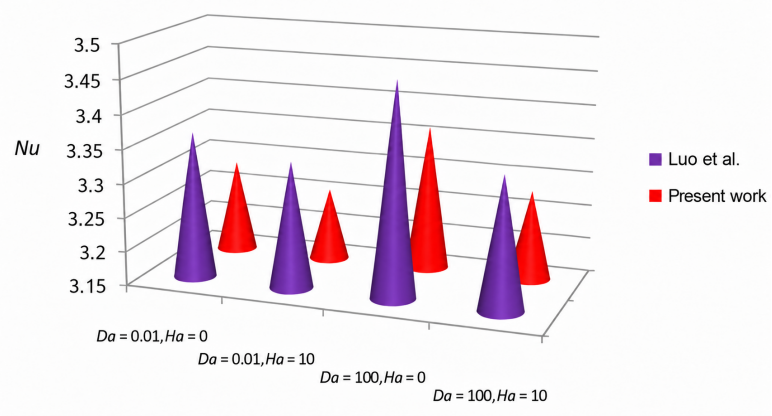



**Figure 3.** Validation of the numerical model through comparison of the average $Nu$ with the reference data reported by Luo et al. [27]

Note: $Nu$ = Nusselt number; $Da$ = Darcy number; $Ha$ = Hartmann number.

**Table 2.** Grid-independence assessment based on the average Nusselt number at $Ra = 10^5$, $Da = 100$, $Ha = 10$, and $\varphi = 0.04$

| Computational Grid Size | Average Nusselt Number |
|---|---|
| $41 \times 121$ | 9.241356 |
| $51 \times 151$ | 9.259779 |
| $61 \times 181$ | 9.269309 |
| $71 \times 211$ | 9.269544 |
| $81 \times 241$ | 9.269803 |

Note: $Ra$ = Rayleigh number; $Da$ = Darcy number; $Ha$ = Hartmann number; $\varphi$ = nanoparticle volume fraction.

Changes in the isotherm patterns provided further evidence of the influence of magnetic forcing. As $Ha$ increased, the temperature contours became less strongly distorted by fluid circulation and moved toward a distribution associated with weaker convection. The reduction in convective mixing limited the redistribution of thermal energy by fluid motion, allowing conductive and modeled radiative transport to make a relatively greater contribution to the thermal field. The magnetic field therefore acted primarily as a regulator of convective intensity rather than as a mechanism that increased the overall heat transfer rate under the investigated conditions. For thermal management applications, this distinction is important because the required response may involve limiting heat transport or controlling temperature distributions rather than maximizing convection under every operating condition.

In contrast, an increase in $Ra$ strengthened the buoyancy-driven motion. The larger thermal driving force promoted more vigorous circulation and increased the deformation of the temperature contours. Thermal plumes developed from the heated boundary and interacted with the surrounding flow, producing more pronounced convection patterns. The resulting circulation facilitated the redistribution of thermal energy within the enclosure and, under the prescribed heating conditions, was associated with lower temperatures along the heated surface. The influence of buoyancy became particularly evident when magnetic damping was absent or relatively weak. These results indicate that the effectiveness of magnetic regulation depends on the underlying strength of natural convection, since the thermal response to the imposed field changes with the buoyancy conditions.

Porous-medium permeability also affected the circulation structure. Increasing the $Da$ reduced the resistance imposed by the porous matrix and allowed the fluid to move more readily through the enclosure. The resulting streamline patterns exhibited stronger recirculation, while the isotherms became more distorted as convective transport intensified. Conversely, lower $Da$ restricted fluid motion and reduced the contribution of convection to heat redistribution. The interaction between permeability and magnetic forcing is especially relevant because the two mechanisms act through different contributions to momentum transport: increasing permeability reduces porous resistance, whereas the electromagnetic terms can suppress circulation. Consequently, the resulting thermal response depends on their combined influence rather than on either parameter alone.

The quantitative effects of the $Ha$, $Ra$, and $Da$ on heat transfer are compared in Figure 6. Increasing the $Da$ raised the average Nusselt number by approximately 4.58% when $Ha$ = 0 and by 3.48% when $Ha$ = 10. These increases were associated with reduced porous resistance and stronger fluid circulation. Although higher

permeability facilitated convection in both cases, its influence was less pronounced in the presence of magnetic forcing. This difference indicates that the electromagnetic damping represented in the model limited part of the convective response produced by the reduction in porous resistance.

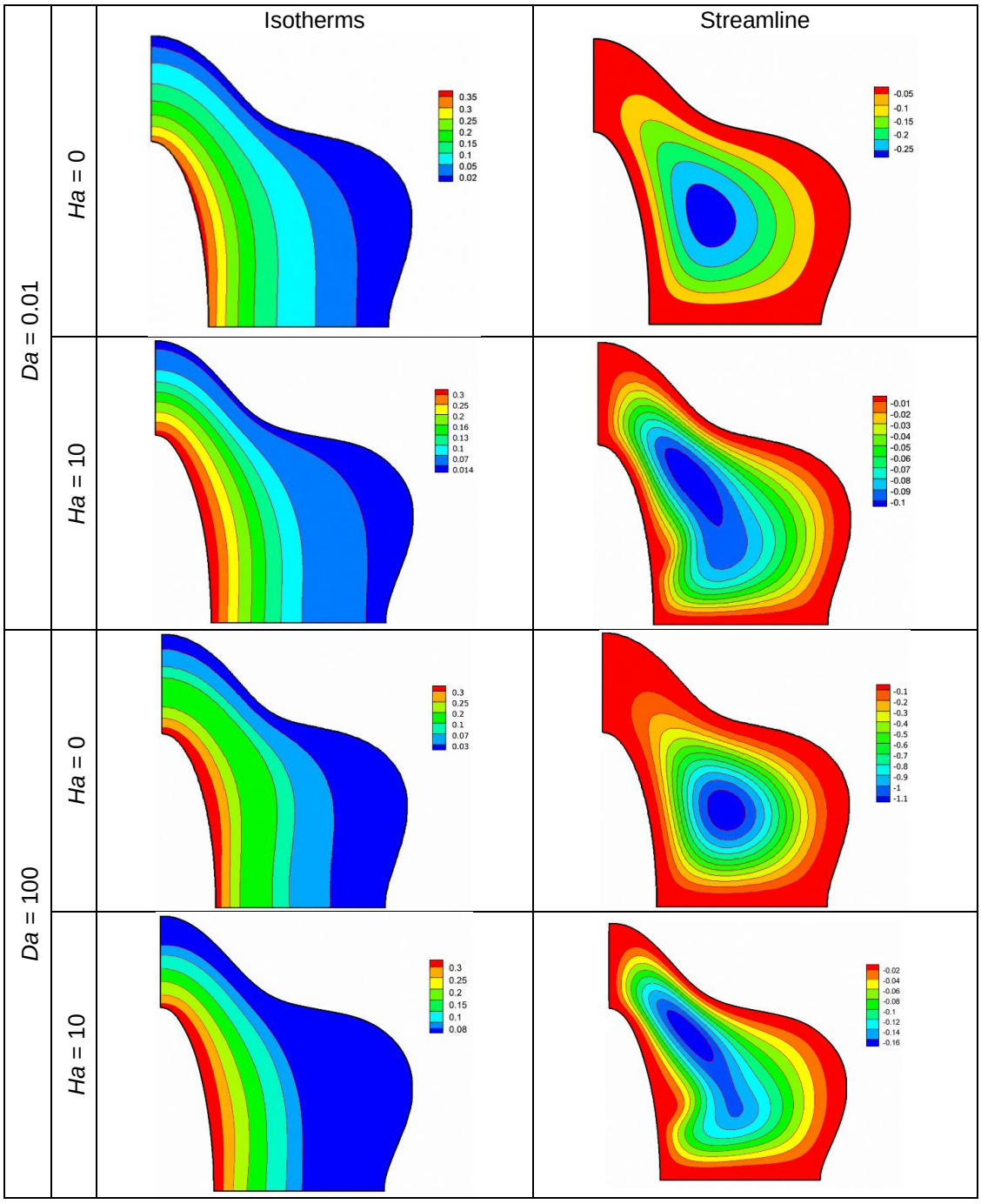



**Figure 4.** Streamline and isotherm distributions for different $Ha$ and $Da$ at $Ra$ = $10^3$ and $Rd$ = 0.8

Note: $Ha$ = Hartmann number; $Ra$ = Rayleigh number; $Da$ = Darcy number; $Rd$ = radiation parameter.

$Ra$ had a substantially greater influence on the average Nusselt number over the investigated range. In the absence of magnetic forcing, increasing $Ra$ produced increases of approximately 149.42% and 150.22% for $Da$ = 0.01 and $Da$ = 100, respectively. When $Ha$ = 10, the corresponding increases were reduced to approximately 79.41% and 84.76%. Thus, the effect of increasing buoyancy remained positive under magnetic forcing, but the magnitude of the resulting heat transfer increase was considerably smaller. The results suggest that the applied magnetic field

became an important constraint on buoyancy-driven circulation, particularly when the flow would otherwise have been intensified by a larger $Ra$.

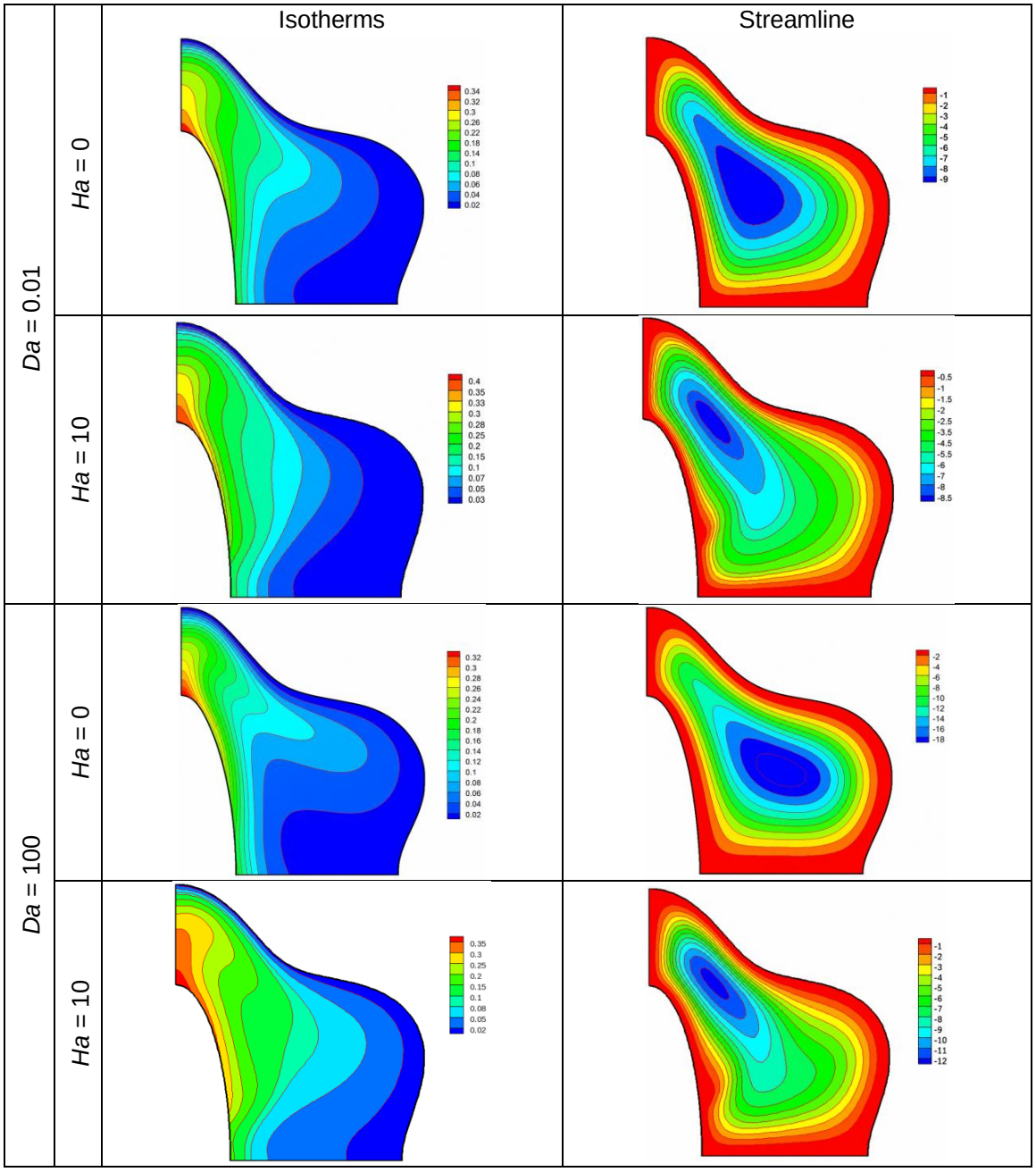



**Figure 5.** Streamline and isotherm distributions for different $Ha$ and $Da$ at $Ra$ = $10^5$ and $Rd$ = 0.8

Note: $Ha$ = Hartmann number; $Ra$ = Rayleigh number; $Da$ = Darcy number; $Rd$ = radiation parameter.

The direct influence of magnetic forcing was reflected in the reduction in average Nusselt number with increasing $Ha$. The calculated reductions reached approximately 28.83% for $Da$ = 0.01 and 29.72% for $Da$ = 100. These values demonstrate that magnetic damping had a substantial influence on the thermal response of the enclosure. The slightly greater reduction at higher permeability suggests that the magnetic contribution became more apparent when the porous matrix imposed less resistance to fluid movement. In a more permeable enclosure, natural convection can develop more readily, leaving a greater contribution from fluid circulation to be restricted by the applied magnetic field. The relationship between magnetic forcing and porous resistance is therefore important when considering the controllability of heat transfer in confined ferrofluid systems.

The contrasting effects of buoyancy, porous resistance, and magnetic forcing indicate that the thermal response

of the enclosure can be adjusted through the governing flow conditions. Stronger buoyancy and greater permeability promoted convective heat transfer, whereas increasing magnetic forcing restricted fluid circulation. This distinction is relevant to thermal components operating under different heat transfer requirements, although the practical effectiveness of magnetic control also depends on the energy required to generate the applied field.

The influence of thermal radiation on the flow and temperature fields is presented in Figure 7, while Figure 8 shows the corresponding variation in average Nusselt number with $Rd$. Increasing $Rd$ modified the temperature distribution and increased the contribution of radiative transport represented in the energy equation. The comparison of the temperature contours indicated that radiation changed the spatial distribution of thermal energy within the enclosure, while the streamline patterns reflected the accompanying adjustment of buoyancy-driven circulation. The resulting thermal response was determined by the interaction between the radiation term and the convective transport mechanisms rather than by radiation acting independently of the flow field.

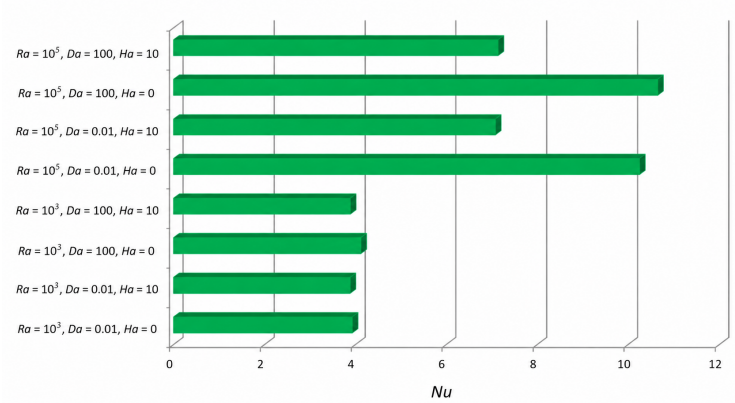



**Figure 6.** Effects of the $Da$, $Ha$, and $Ra$ on the average $Nu$ at $\varphi = 0.04$

Note: $Ha$ = Hartmann number; $Ra$ = Rayleigh number; $Da$ = Darcy number; $Nu$ = Nusselt number.

The calculated average Nusselt number increased by approximately 104.75% when $Ha = 0$ and by 94.63% when $Ha = 10$ as $Rd$ increased over the investigated range. These results show that radiation had a pronounced influence on the overall heat transfer indicator adopted in the numerical formulation. The smaller increase under magnetic forcing indicates that the thermal response was modified by the suppression of buoyancy-driven circulation. The difference between the two magnetic conditions also shows that the effects of thermal radiation and magnetic damping were not independent within the modeled enclosure.

The interpretation of these results requires consideration of the adopted Nusselt number definition. Because the formulation incorporates a radiative contribution, an increase in the reported Nusselt number does not necessarily correspond to an equivalent increase in convection alone. Instead, it reflects the combined thermal response represented by the governing energy equation and the heat transfer expression. This distinction is particularly important for assessing energy-related applications, since changes in a dimensionless heat transfer indicator do not directly establish improvements in device efficiency or reductions in energy consumption.

The pronounced response to $Rd$ indicates that radiative transport should be considered when evaluating the thermal behavior of enclosures operating under conditions where radiation is appreciable. However, the reported increase in the average Nusselt number reflects the adopted heat transfer formulation and does not independently establish a corresponding improvement in device-level thermal efficiency.

The effect of $Fe_3O_4$ nanoparticle concentration on the average Nusselt number is shown in Figure 9. Increasing $\varphi$ changed the effective thermophysical properties of the working fluid, particularly its thermal conductivity and viscosity, and produced a moderate increase in the calculated heat transfer indicator. The average Nusselt number increased by approximately 5.98% when $Ha = 0$ and by 3.70% when $Ha = 10$. The smaller increase under magnetic forcing was consistent with the reduction in fluid circulation observed in the earlier parameter studies. Although the addition of $Fe_3O_4$ nanoparticles increased the effective thermal conductivity in the adopted property model, the resulting heat transfer response remained dependent on the interaction between thermal diffusion, viscous effects, and convection.

The relatively modest changes associated with nanoparticle loading, compared with those obtained by varying $Ra$ or $Rd$, indicate that increasing nanoparticle concentration was not the dominant means of modifying the thermal response within the investigated parameter ranges. This observation is relevant when considering ferrofluid-based thermal management because higher particle loading may introduce additional practical constraints, including increased viscosity, suspension stability requirements, and material costs. The numerical model does not quantify these factors, but the results suggest that nanoparticle concentration should be evaluated alongside the operating

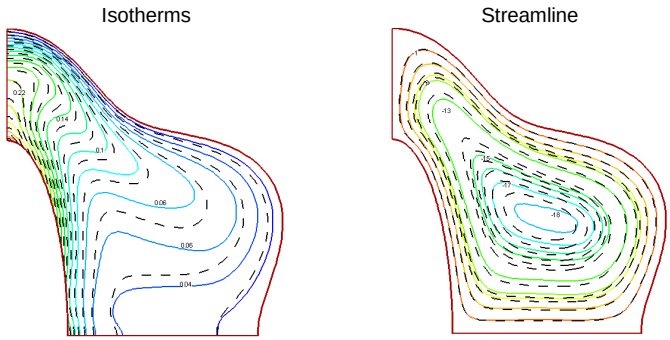



**Figure 7.** Effects of thermal radiation on the streamline and isotherm distributions at $Da$ = 100, $Ra$ = $10^5$, and $\varphi$ = 0.04

Note: Solid and dashed contours correspond to $Rd$ = 0 and $Rd$ = 0.8, respectively. $Ra$ = Rayleigh number; $Da$ = Darcy number; $Rd$ = radiation parameter ; $\varphi$ = nanoparticle volume fraction.

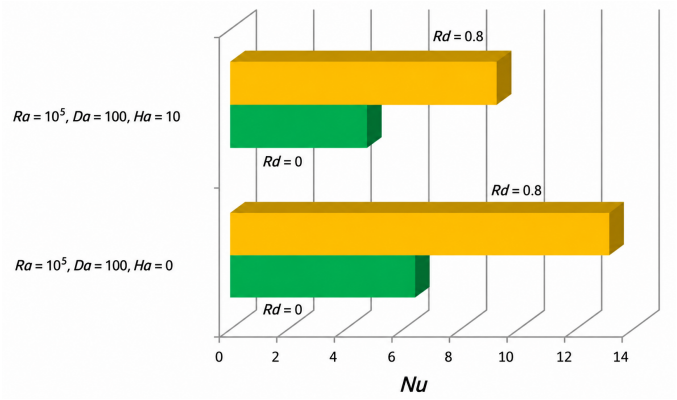



**Figure 8.** Variation in the average $Nu$ with $Rd$ under different magnetic conditions

Note: $Ha$ = Hartmann number; $Ra$ = Rayleigh number; $Da$ = Darcy number; $Rd$ = radiation parameter ; $Nu$ = Nusselt number.

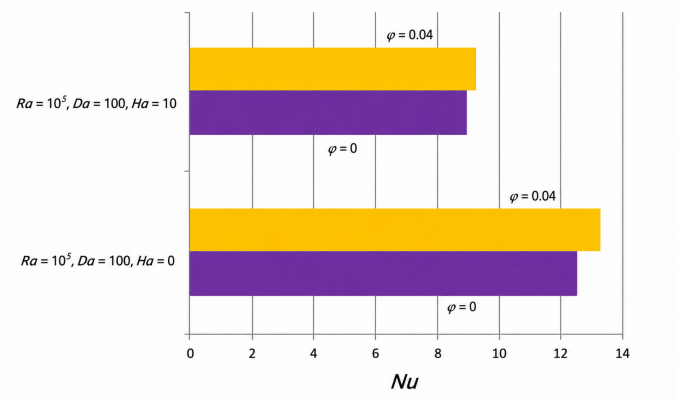



**Figure 9.** Effect of the $Fe_3O_4$ $\varphi$ on the average $Nu$ under different magnetic conditions

Note: $Ha$ = Hartmann number; $Ra$ = Rayleigh number; $Da$ = Darcy number; $\varphi$ = nanoparticle volume fraction; $Nu$ = Nusselt number.

conditions and magnetic control strategy rather than treated as an independent route to improved thermal performance.

Considering the parameter effects together, the results reveal different degrees of sensitivity to the mechanisms governing heat transfer. Buoyancy and thermal radiation produced the largest reported increases in average Nusselt number over their respective investigated ranges, while magnetic forcing caused a substantial reduction. Increasing porous-medium permeability and nanoparticle concentration produced smaller positive changes. These percentages correspond to different parameter variations and operating conditions; they therefore should not be interpreted as a normalized ranking of parameter importance. Nevertheless, the comparison illustrates that adjusting a single

material or geometric property may have a more limited influence than changing the thermal driving conditions or the modeled radiation contribution.

For sustainable energy thermal management, the findings suggest that magnetic forcing and passive material properties can be considered together when assessing the thermal response of confined ferrofluid systems. Porous-medium permeability and nanoparticle loading affect the fluid and transport characteristics, while an externally applied magnetic field provides an additional means of regulating circulation. These interactions are potentially relevant to solar thermal components, heat exchangers, and thermal storage devices where controlled heat transfer is required. Their practical value, however, must be established through component-level analysis that accounts for operating conditions and the energy consumed by the magnetic source.

The present findings should, however, be interpreted within the limitations of the numerical model. The calculations were performed for an idealized two-dimensional enclosure using a homogeneous nanofluid approximation and prescribed electromagnetic and radiative formulations. The model did not resolve particle aggregation, sedimentation, detailed magnetic microstructure, or the full operating characteristics of an energy conversion or storage device. Furthermore, the analysis did not quantify the electrical power required to generate the magnetic field, pumping requirements, entropy generation, exergy efficiency, or lifecycle environmental effects. Consequently, the reported changes in Nusselt number cannot be translated directly into energy savings, improved sustainability, or reduced emissions.

Within these limits, the results provide a quantitative description of how magnetic forcing, buoyancy, porous permeability, radiation, and nanoparticle concentration interact in a complex ferrofluid enclosure. The findings identify conditions under which the modeled magnetic field suppresses convection and show how other transport mechanisms modify the overall thermal response. This understanding provides a starting point for further numerical and experimental investigations of controllable heat transfer in energy-related thermal components, where both thermal performance and the energy required for external regulation must be considered.

## 4 Conclusions

This study investigated the coupled effects of magnetic forcing, buoyancy, porous-medium permeability, thermal radiation, and nanoparticle concentration on natural convection in a geometrically complex enclosure filled with $Fe_3O_4$–water ferrofluid. The numerical analysis was conducted using the CVFEM, considering an inner heated elliptical boundary, an outer sinusoidal cooled wall, and a spatially non-uniform magnetic field. The results revealed distinct interactions among the governing transport mechanisms and their influence on fluid circulation and heat transfer. The principal findings are summarized below.

(1) Increasing $Ha$ suppressed natural convection and reduced the average Nusselt number by approximately 28.83% and 29.72% at $Da$ of 0.01 and 100, respectively. The reduction was associated with weaker circulation and changes in the temperature distribution. Within the adopted numerical model, magnetic forcing therefore acted as a means of restricting convective transport, although its usefulness for thermal management depends on the intended operating conditions.

(2) Buoyancy had a pronounced influence on the thermal response. In the absence of magnetic forcing, increasing $Ra$ raised the average Nusselt number by approximately 149.42% and 150.22% at $Da$ of 0.01 and 100, respectively. When $Ha$ was 10, the corresponding increases were reduced to 79.41% and 84.76%. These differences indicate that the magnetic contribution substantially limited the increase in convective heat transfer produced by stronger buoyancy.

(3) Increasing porous-medium permeability promoted fluid circulation and produced moderate improvements in the calculated heat transfer indicator. The average Nusselt number increased by approximately 4.58% without magnetic forcing and 3.48% at $Ha$ of 10. The smaller increase under magnetic forcing suggests that the reduction in porous resistance did not fully offset the damping of buoyancy-driven circulation.

(4) Thermal radiation had a considerable effect on the overall thermal response. Increasing $Rd$ raised the average Nusselt number by approximately 104.75% and 94.63% at $Ha$ of 0 and 10, respectively. These increases reflect the influence of the radiative contribution included in the numerical formulation and should not be interpreted solely as increases in convective heat transfer. The results also indicate that magnetic forcing modified the thermal response associated with radiation.

(5) Increasing the $Fe_3O_4$ $\varphi$ raised the average Nusselt number by approximately 5.98% in the absence of magnetic forcing and 3.70% at $Ha$ of 10. Compared with the effects of buoyancy and radiation over the investigated parameter ranges, nanoparticle loading produced a relatively modest change in the calculated heat transfer indicator. Its influence was governed by the combined changes in effective thermophysical properties and fluid circulation.

Taken together, these findings show that magnetic forcing can regulate convection in a porous ferrofluid enclosure, while permeability, radiation, buoyancy, and nanoparticle loading determine the resulting thermal response. This interaction is relevant to the investigation of controllable thermal components for sustainable energy technologies, including solar thermal systems, heat exchangers, and thermal energy storage arrangements. In such applications, the

desired thermal behavior may involve either promoting heat removal or restricting unwanted heat transfer, depending on the operating requirements. The present results provide a numerical basis for examining these competing objectives but do not establish improvements in energy efficiency at the system level.

The study was limited to an idealized two-dimensional enclosure, a homogeneous ferrofluid representation, and the prescribed electromagnetic and radiative models. The energy required to generate the magnetic field, nanoparticle stability, entropy generation, and exergy efficiency were not evaluated. Further investigations should therefore consider experimentally supported magnetic properties, more realistic thermal boundary conditions, and complete energy balances. Extending the analysis to component-level configurations would help determine whether the observed control of convective transport can contribute to improved thermal management without introducing excessive energy consumption or other operational penalties.

## Author Contributions

Conceptualization, M.J.S. and M.R.D.; methodology, S.M.E.; software, A.Y.; validation, M.J.S., M.R.D., and A.Y.; formal analysis, S.M.E.; investigation, M.J.S.; resources, M.R.D.; data curation, A.Y.; writing—original draft preparation, M.J.S.; writing—review and editing, M.J.S., M.R.D., S.M.E., and A.Y.; visualization, M.J.S.; supervision, A.Y. All authors have read and agreed to the published version of the manuscript.

## Data Availability

The data used to support the research findings are available from the corresponding author upon request.

## Conflicts of Interest

The authors declare no conflicts of interest.

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